

Chapter 22. Curl, Divergence, and Surface Integrals

Local rotation, local expansion, oriented surfaces, and flux

The previous chapters developed line integrals and Green's theorem. A line integral measures how a scalar or vector field accumulates along a curve. Green's theorem showed that circulation or flux around a closed plane curve can be computed by integrating a local derivative over the enclosed region.

This chapter moves that idea into three-dimensional space. Instead of curves enclosing planar regions, we now study surfaces in space. Instead of asking only how much a vector field pushes along a path, we ask how much it rotates near a point, how much it expands out of a point, and how much it flows through a surface.

The central objects are curl, divergence, scalar surface integrals, and flux integrals. They prepare the way for Stokes' theorem and the divergence theorem.

Learning goals

After studying this chapter, you should be able to:

1. compute the curl of a vector field in space;
2. interpret curl as local rotational tendency;
3. compute and interpret divergence;
4. use the identities $\nabla \cdot (\nabla \times F) = 0$ and $\nabla \times (\nabla f) = 0$ when appropriate;
5. parameterize surfaces and compute tangent vectors;
6. compute surface area using $|r_u \times r_v|$;
7. compute scalar surface integrals;
8. orient a surface and compute flux integrals;
9. use graph formulas for surface area and flux;
10. approximate curl, divergence, surface area, and flux numerically.

The main theme

Curl and divergence are local differential measurements. Surface integrals and flux integrals are global accumulations. The great theorems of vector calculus explain when local rotation or local expansion determines a global boundary integral.

1 The differential operator ∇

For a scalar function $f(x, y, z)$, the gradient is

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

For a vector field

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle,$$

we use the formal differential operator

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This notation organizes three operations:

$$\nabla f \quad \text{gradient}, \quad \nabla \cdot F \quad \text{divergence}, \quad \nabla \times F \quad \text{curl}.$$

The notation is formal: ∇ is not an ordinary vector of numbers. Its components are derivative operators.

2 Curl

Let $F = \langle P, Q, R \rangle$ be a vector field in space. The *curl* of F is

$$\nabla \times F = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle.$$

Equivalently,

$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

Curl measures the infinitesimal tendency of the field to rotate. Its direction is the axis of rotation determined by the right-hand rule, and its magnitude measures the local strength of rotation.

Curl in the plane

A two-dimensional vector field $F(x, y) = \langle P(x, y), Q(x, y) \rangle$ can be treated as the three-dimensional field $\langle P, Q, 0 \rangle$. Then

$$\nabla \times F = \langle 0, 0, Q_x - P_y \rangle.$$

The scalar quantity $Q_x - P_y$ is the scalar curl of the plane field. It is exactly the integrand in Green's theorem in circulation form.

Example 2.1: A rotating field

Let

$$F(x, y, z) = \langle -y, x, 0 \rangle.$$

Then $P = -y$, $Q = x$, and $R = 0$. Hence

$$\nabla \times F = \langle 0 - 0, 0 - 0, 1 - (-1) \rangle = \langle 0, 0, 2 \rangle.$$

The field rotates counterclockwise in the xy -plane. The curl points in the positive z -direction.

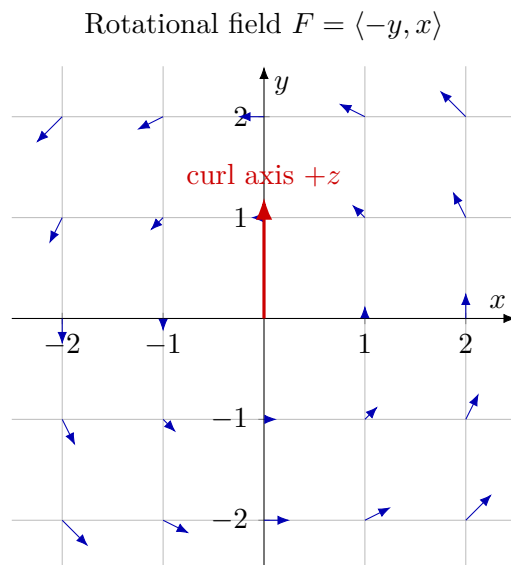


Figure 1: The field $F = \langle -y, x \rangle$ rotates counterclockwise. Its curl points in the positive z -direction.

Example 2.2: Curl of a three-dimensional field

Let

$$F(x, y, z) = \langle yz, xz, xy + e^z \rangle.$$

Then $P = yz$, $Q = xz$, and $R = xy + e^z$. We have

$$R_y = x, \quad Q_z = x, \quad P_z = y, \quad R_x = y, \quad Q_x = z, \quad P_y = z.$$

Hence

$$\nabla \times F = \langle x - x, y - y, z - z \rangle = \langle 0, 0, 0 \rangle.$$

Indeed, $F = \nabla f$ for $f(x, y, z) = xyz + e^z$.

3 Divergence

Let $F = \langle P, Q, R \rangle$. The *divergence* of F is the scalar field

$$\nabla \cdot F = P_x + Q_y + R_z.$$

Divergence measures the local tendency of a vector field to flow outward from a point. Positive divergence indicates source-like behavior, negative divergence indicates sink-like behavior, and zero divergence indicates local incompressibility.

Divergence is a scalar field

Curl returns a vector field in three dimensions. Divergence returns a scalar field:

$$\nabla \times F \in \mathbb{R}^3, \quad \nabla \cdot F \in \mathbb{R}.$$

Example 3.1: A source field and an incompressible rotational field

For $F(x, y, z) = \langle x, y, z \rangle$,

$$\nabla \cdot F = 1 + 1 + 1 = 3.$$

The field points away from the origin and has positive divergence.

For $F(x, y, z) = \langle -y, x, 0 \rangle$,

$$\nabla \cdot F = 0 + 0 + 0 = 0.$$

The field rotates, but it does not locally expand or compress volume.

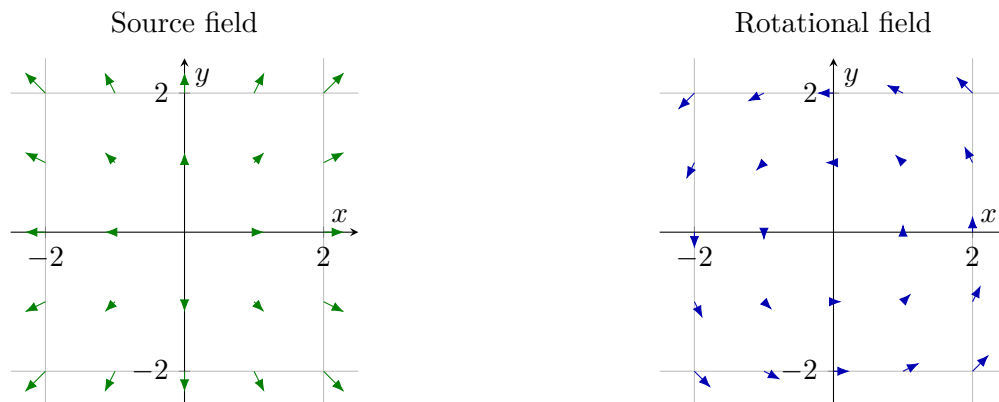


Figure 2: A radial source field has positive divergence. A rotational field can have zero divergence but nonzero curl.

4 Two fundamental identities

For sufficiently smooth functions and fields, two identities occur throughout vector calculus:

$$\nabla \times (\nabla f) = 0, \quad \nabla \cdot (\nabla \times F) = 0.$$

The first says that gradient fields have no local rotation. It is the three-dimensional version of the conservative-field test. The second says that a curl field has no local source or sink.

Smoothness and domain matter

These identities require enough differentiability. Also, equations such as $\nabla \times F = 0$ guarantee a global potential only under suitable assumptions on the domain. Holes and singularities can create global circulation even when local curl is zero away from the singularity.

5 Numerical curl and divergence on a grid

In applications, a vector field may be sampled rather than given by a formula. For a plane field $F = \langle P, Q \rangle$, finite differences approximate

$$\nabla \cdot F = P_x + Q_y, \quad \text{curl}_z F = Q_x - P_y.$$

This is used for fluid simulation, weather models, image motion, and learned dynamical systems.

A sampled field with divergence and scalar curl

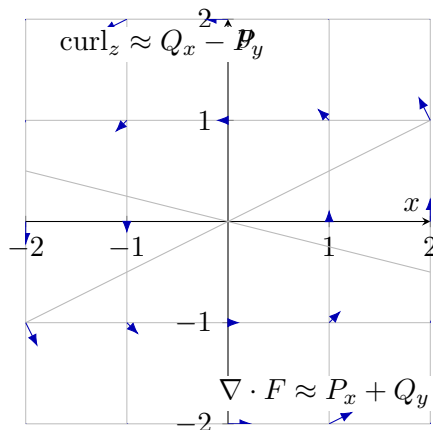


Figure 3: Finite differences approximate divergence and scalar curl from neighboring values on a grid.

6 Surfaces and parameterizations

A curve is described by one parameter:

$$r(t) = \langle x(t), y(t), z(t) \rangle.$$

A surface is described by two parameters:

$$r(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

The parameter domain D lies in the uv -plane. The map r bends and stretches this flat parameter region into a surface S in space.

For a parameterized surface, the tangent vectors are

$$r_u = \frac{\partial r}{\partial u}, \quad r_v = \frac{\partial r}{\partial v}.$$

The cross product $r_u \times r_v$ is normal to the surface. Its magnitude gives the local area scaling factor:

$$dS = |r_u \times r_v| du dv.$$

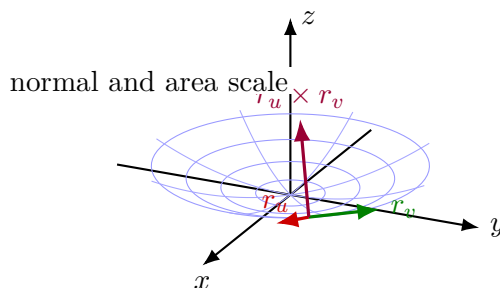


Figure 4: For a parameterized surface, $r_u \times r_v$ gives a normal direction and the local area scale.

7 Surface area

The area of a parameterized surface $r(u, v)$, $(u, v) \in D$, is

$$A(S) = \iint_D |r_u \times r_v| dA.$$

This is the surface analog of arc length. For curves, $|r'(t)|$ measures how much the parameter stretches length. For surfaces, $|r_u \times r_v|$ measures how much the two-parameter map stretches area.

Graph surfaces

If a surface is a graph $z = g(x, y)$ over D , parameterize it as

$$r(x, y) = \langle x, y, g(x, y) \rangle.$$

Then

$$r_x = \langle 1, 0, g_x \rangle, \quad r_y = \langle 0, 1, g_y \rangle,$$

so

$$r_x \times r_y = \langle -g_x, -g_y, 1 \rangle.$$

Therefore

$$dS = \sqrt{1 + g_x^2 + g_y^2} dA, \quad A(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} dA.$$

Example 7.1: Area of a plane patch

Find the area of the part of the plane

$$z = 8 + 2x + 5y$$

above the rectangle $0 \leq x \leq 3$, $1 \leq y \leq 6$. Here $g_x = 2$ and $g_y = 5$, so

$$\sqrt{1 + g_x^2 + g_y^2} = \sqrt{1 + 4 + 25} = \sqrt{30}.$$

The rectangle has area 15, so

$$A(S) = 15\sqrt{30}.$$

8 Scalar surface integrals

If $f(x, y, z)$ is a scalar function defined on a surface S , then the scalar surface integral is

$$\iint_S f(x, y, z) dS.$$

For a parameterized surface $r(u, v)$,

$$\iint_S f dS = \iint_D f(r(u, v)) |r_u \times r_v| du dv.$$

If f is a surface density, this integral gives total mass. If $f = 1$, it gives surface area.

Example 8.1: Surface integral over a paraboloid

Let S be the graph $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 1$. Compute

$$\iint_S z \, dS.$$

Since $g(x, y) = x^2 + y^2$, we have $g_x = 2x$ and $g_y = 2y$, so

$$dS = \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

The integral becomes

$$\iint_D (x^2 + y^2) \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

In polar coordinates,

$$\iint_S z \, dS = \int_0^{2\pi} \int_0^1 r^2 \sqrt{1 + 4r^2} \, r \, dr \, d\theta = 2\pi \int_0^1 r^3 \sqrt{1 + 4r^2} \, dr.$$

Paraboloid over the unit disk

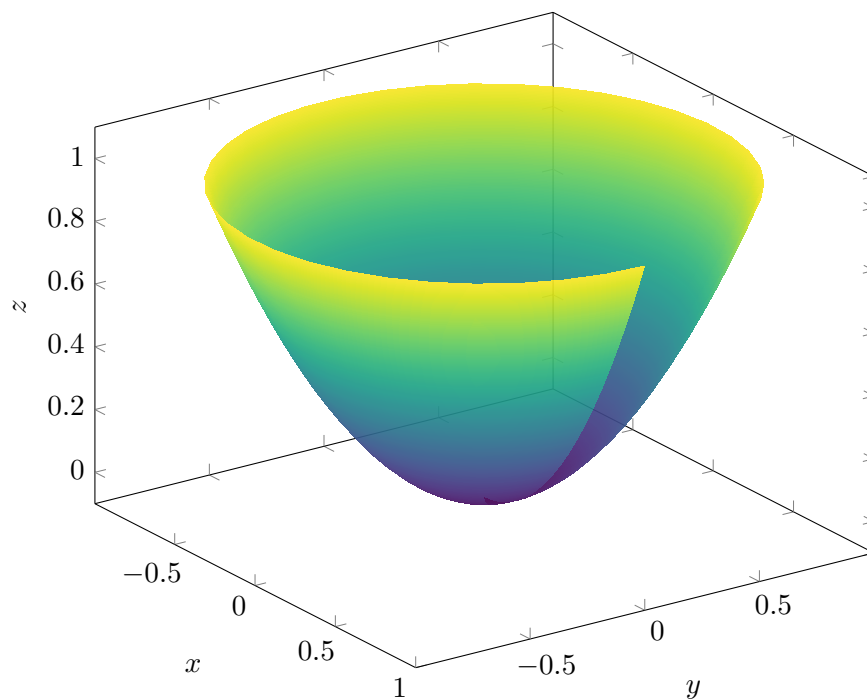


Figure 5: A scalar surface integral accumulates density over a curved surface. Here the density is $f(x, y, z) = z$ on $z = x^2 + y^2$.

9 Orientation of surfaces

A surface is *orientable* if it is possible to choose a continuous unit normal vector n at every point of the surface. Choosing one of the two possible normal directions gives an *orientation*.

For a parameterized surface $r(u, v)$, the vector $r_u \times r_v$ defines one orientation. Reversing the parameter order changes the normal direction:

$$r_v \times r_u = -(r_u \times r_v).$$

For a closed surface, the standard orientation is outward. For an open surface, the orientation must be specified, for example upward, downward, outward, or inward.

Orientation affects flux

Scalar surface integrals do not depend on orientation because dS is positive. Flux integrals do depend on orientation because changing n to $-n$ changes the sign of $F \cdot n$.

10 Flux integrals

Let F be a vector field and let S be an oriented surface with unit normal vector n . The *flux* of F through S is

$$\iint_S F \cdot n \, dS.$$

This measures the net amount of the field passing through the surface in the chosen normal direction.

Using a parameterization $r(u, v)$, we have

$$n \, dS = (r_u \times r_v) \, du \, dv$$

for the orientation determined by $r_u \times r_v$. Therefore

$$\iint_S F \cdot n \, dS = \iint_D F(r(u, v)) \cdot (r_u \times r_v) \, du \, dv.$$

Graph formula for flux

Suppose $S : z = g(x, y)$ over D , with upward orientation. Then

$$r_x \times r_y = \langle -g_x, -g_y, 1 \rangle.$$

If $F = \langle P, Q, R \rangle$, then

$$\iint_S F \cdot n \, dS = \iint_D (-Pg_x - Qg_y + R) \, dA,$$

where P, Q, R are evaluated at $(x, y, g(x, y))$. For downward orientation, the sign changes.

Example 10.1: Flux through a graph

Let $F(x, y, z) = \langle x, y, z \rangle$, and let S be the upper paraboloid

$$z = 1 - x^2 - y^2, \quad x^2 + y^2 \leq 1,$$

with upward orientation. Here $g_x = -2x$ and $g_y = -2y$. The upward flux integrand is

$$-Pg_x - Qg_y + R = -x(-2x) - y(-2y) + (1 - x^2 - y^2) = 1 + x^2 + y^2.$$

Therefore

$$\iint_S F \cdot n \, dS = \iint_D (1 + x^2 + y^2) \, dA.$$

In polar coordinates,

$$\int_0^{2\pi} \int_0^1 (1 + r^2) r \, dr \, d\theta = 2\pi \left[\frac{r^2}{2} + \frac{r^4}{4} \right]_0^1 = \frac{3\pi}{2}.$$

Projected upward normals on $z = 1 - x^2 - y^2$

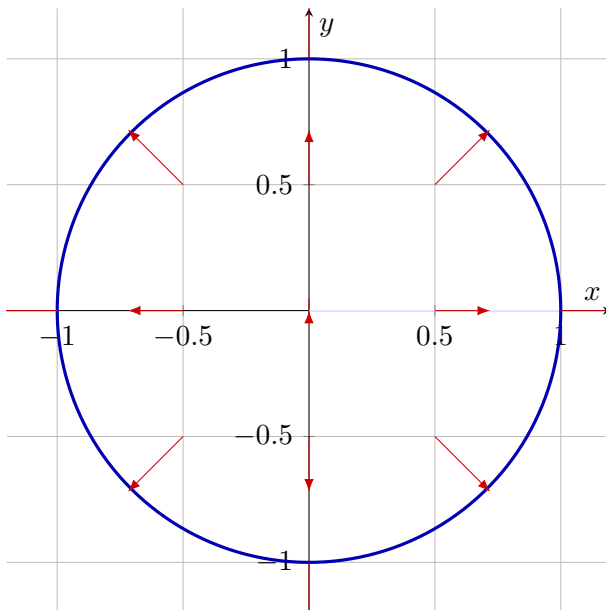


Figure 6: For the upward-oriented graph $z = 1 - x^2 - y^2$, the projected normal arrows in the xy -plane point outward and the normal has positive z -component.

11 Flux through standard surfaces

Sphere

The sphere of radius a centered at the origin can be parameterized by

$$r(\phi, \theta) = \langle a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi \rangle, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

The outward unit normal is

$$n = \frac{1}{a} \langle x, y, z \rangle.$$

If $F = \langle x, y, z \rangle$, then on the sphere,

$$F \cdot n = \langle x, y, z \rangle \cdot \frac{1}{a} \langle x, y, z \rangle = \frac{x^2 + y^2 + z^2}{a} = a.$$

Since the surface area is $4\pi a^2$, the flux is

$$\iint_S F \cdot n \, dS = a(4\pi a^2) = 4\pi a^3.$$

For the sphere $x^2 + y^2 + z^2 = 4$, $a = 2$, so the flux is 32π .

Radial outward normals: cross-section

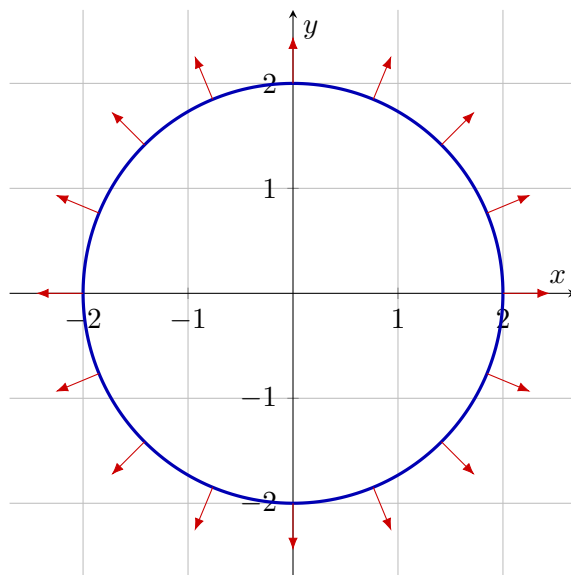


Figure 7: For a sphere centered at the origin, outward normals are radial. The cross-section shows the same orientation idea.

Cylinder

A circular cylinder of radius a can be parameterized by

$$r(\theta, z) = \langle a \cos \theta, a \sin \theta, z \rangle.$$

Then

$$r_\theta = \langle -a \sin \theta, a \cos \theta, 0 \rangle, \quad r_z = \langle 0, 0, 1 \rangle.$$

The cross product

$$r_z \times r_\theta = \langle -a \cos \theta, -a \sin \theta, 0 \rangle$$

points inward, while

$$r_\theta \times r_z = \langle a \cos \theta, a \sin \theta, 0 \rangle$$

points outward. Always check orientation.

12 Numerical surface integration

For computational work, a parameterized surface integral is usually approximated by sampling a grid in parameter space. If $r(u, v)$, $(u, v) \in D$, then

$$\iint_S f \, dS \approx \sum_{i,j} f(r(u_i, v_j)) |r_u(u_i, v_j) \times r_v(u_i, v_j)| \Delta u \Delta v.$$

Similarly,

$$\iint_S F \cdot n \, dS \approx \sum_{i,j} F(r(u_i, v_j)) \cdot (r_u(u_i, v_j) \times r_v(u_i, v_j)) \Delta u \Delta v.$$

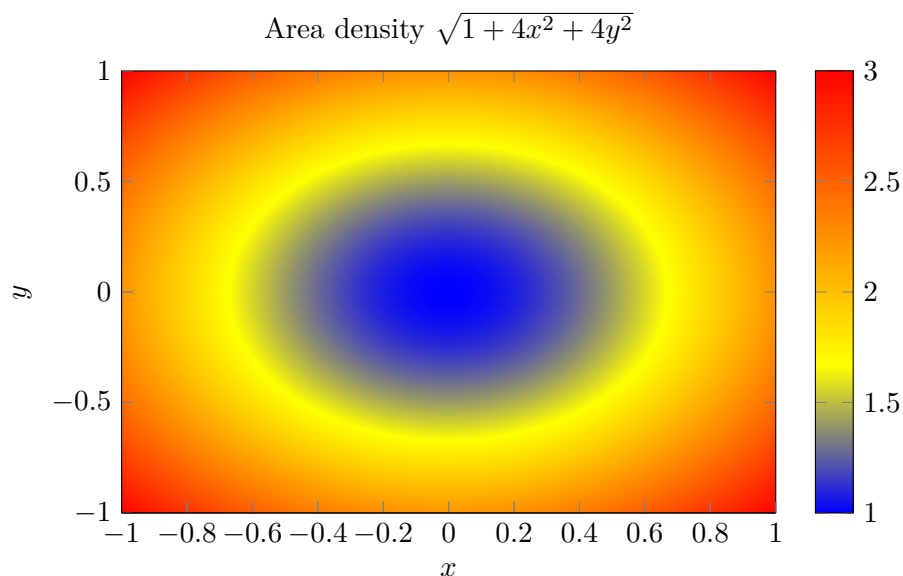


Figure 8: Numerical surface integration samples the parameter domain and multiplies by the local area scale.

The plot shows the area-density factor for $z = x^2 + y^2$ over a square. Summing this density over a fine grid is a Riemann-sum approximation to surface area.

13 Curl, divergence, and flux in applied modeling

Fluid velocity fields

If F is a fluid velocity field, then $\nabla \times F$ measures local swirling, $\nabla \cdot F$ measures local expansion or compression, and $\iint_S F \cdot n \, dS$ measures the rate of flow through a surface. These ideas appear in incompressible flow, fluid simulation, weather models, and diffusion processes.

Probability flow

In probability and statistics, a time-dependent density can move under a velocity field. Divergence helps describe whether probability mass is locally spreading or concentrating. In continuous normalizing flows and score-based generative modeling, vector fields and Jacobians are central tools.

Optimization and machine learning

Gradient descent is driven by a vector field $-\nabla L$. Such a field has zero curl when L is smooth. More general training dynamics may include momentum, stochastic noise, constraints, or game-

theoretic interactions. These dynamics can show rotational behavior, naturally detected by curl-like measurements.

Geometry and data

Surface integrals appear when data lie on curved manifolds rather than flat Euclidean spaces. Area elements and Jacobian factors explain why densities change under parameterization and why sampling uniformly in parameter coordinates does not always sample uniformly on a surface.

14 Common mistakes

1. Forgetting that curl is a vector in space. In three dimensions, $\nabla \times F$ has three components.
2. Confusing scalar curl with full curl. In the plane, $Q_x - P_y$ is the z -component of the three-dimensional curl of $\langle P, Q, 0 \rangle$.
3. Forgetting the area factor. The factor $|r_u \times r_v|$ is essential in a surface integral.
4. Using the wrong orientation. Flux changes sign when the normal direction is reversed.
5. Using graph formulas outside their setting. The formula for $z = g(x, y)$ applies only when the surface is represented as a graph over the xy -plane.
6. Treating dS like dA . For curved surfaces, dS includes an area-stretching factor.

15 Chapter exercises

Conceptual exercises

1. Explain the difference between curl and divergence.
2. Give an example of a vector field with nonzero curl and zero divergence.
3. Give an example of a vector field with zero curl and nonzero divergence.
4. Explain why scalar surface integrals do not depend on orientation but flux integrals do.
5. What does the vector $r_u \times r_v$ represent geometrically?
6. Why does reversing the order of the cross product reverse the orientation?
7. Explain why $\nabla \times (\nabla f) = 0$ should be expected geometrically.
8. Explain why $\nabla \cdot (\nabla \times F) = 0$ suggests that curl fields have no sources or sinks.

Computational exercises

9. Compute the curl of $F(x, y, z) = \langle xz, yz, xy \rangle$.
10. Compute the divergence of $F(x, y, z) = \langle x^2y, y^2z, z^2x \rangle$.
11. Determine whether $F(x, y, z) = \langle yz, xz, xy \rangle$ has zero curl.

12. Let $F(x, y, z) = \langle -y, x, z \rangle$. Compute $\nabla \times F$ and $\nabla \cdot F$.
13. Let S be the graph $z = x^2 + y^2$ over $0 \leq x \leq 1$, $0 \leq y \leq 1$. Set up the surface area integral.
14. Compute the area of the plane $z = 1 + 2x - 2y$ above the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$.
15. Let S be the sphere of radius 3 centered at the origin. Compute the outward flux of $F = \langle x, y, z \rangle$ through S .
16. Let S be the graph $z = 4 - x - y$ over the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Compute the upward flux of $F = \langle 0, 0, z \rangle$ through S .
17. Parameterize the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$, and compute $|r_\theta \times r_z|$.
18. Let $r(u, v) = \langle u, v, u^2 - v^2 \rangle$, $0 \leq u \leq 1$, $0 \leq v \leq 1$. Compute $r_u \times r_v$ and set up the surface area integral.

Applied and computational exercises

19. Approximate the surface area of $z = \sin(xy)$ over $[-1, 1] \times [-1, 1]$.
20. Approximate the flux of $F = \langle x, y, z \rangle$ through the upward-oriented graph $z = 1 - x^2 - y^2$ over the unit disk.
21. Suppose a measured two-dimensional velocity field has approximately zero divergence but nonzero scalar curl. Give an applied interpretation.
22. Suppose a learned dynamical system in parameter space has visible rotational motion around a point. Which derivative measurement from this chapter would detect that behavior?
23. A surface is parameterized by a neural network $r(u, v)$. Explain why $|r_u \times r_v|$ is the correct local area distortion factor.
24. In a probability flow model, why is divergence related to local density change?

16 Chapter summary

This chapter introduced the local derivative operations and surface integrals needed for the final theorems of vector calculus.

- Curl measures local rotation.
- Divergence measures local source or sink behavior.
- In the plane, scalar curl is $Q_x - P_y$.
- In space, curl is $\nabla \times F$ and divergence is $\nabla \cdot F$.
- The identity $\nabla \times (\nabla f) = 0$ says gradient fields have no local rotation.
- The identity $\nabla \cdot (\nabla \times F) = 0$ says curl fields have no local source or sink.
- A parameterized surface $r(u, v)$ has tangent vectors r_u and r_v .
- The vector $r_u \times r_v$ gives a normal direction and an area scale.

- Scalar surface integrals accumulate scalar quantities over curved surfaces.
- Flux integrals measure how much a vector field passes through an oriented surface.
- Orientation matters for flux but not for scalar surface integrals.

The next chapter uses these tools to state and apply Stokes' theorem and the divergence theorem, the three-dimensional successors of Green's theorem.